

ADARSH KUMAR SINGH

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SUMMARY

Results-driven Full Stack AI Software Engineer specializing in Large Language Models (LLMs), Agentic Workflows, and scalable Python architectures. Proven ability to engineer autonomous multi-agent systems, implement Retrieval-Augmented Generation (RAG) pipelines, and build robust enterprise LLM guardrails from the ground up. Seeking an AI Engineering role to leverage expertise in Python, FastAPI, and GenAI to build secure, highly automated applications.

SKILLS

Programming Languages: Python, JavaScript, SQL, Bash

AI & Machine Learning: Large Language Models (LLaMA 3, GPT), Retrieval-Augmented Generation (RAG), Multi-Agent Orchestration (ReAct Loops), AI Guardrails, Prompt Engineering

Frameworks & Libraries: FastAPI, Pytest, Sentence-Transformers, ChromaDB (Vector Databases), GSAP, Three.js

Cloud & Infrastructure: Oracle Cloud Infrastructure (OCI), Oracle Database Design, Data Warehousing, Server-Sent Events (SSE)

Developer Tools: Git, REST APIs, JSON Validation, CI/CD Pipelines

PROJECTS

Autonomous Self-Healing DevOps Swarm

Multi-Agent Orchestration | Python, LLaMA 3, Pytest, GitHub API | Mar 2026 - Present

- Engineered a 4-agent orchestration framework using ReAct loops to intercept CI/CD pipeline failures, automatically diagnose root cause, generate patches, and open PRs, reducing mean-time-to-recovery (MTTR) by 90% across 50+ test scenarios.
- Achieved 85% first-pass fix accuracy on Python runtime errors by implementing a continuous diagnostic loop with LLaMA 3, sandbox execution, and automated Pytest verification before PR submission.
- Designed dynamic agent routing (Diagnoser > Developer > Verifier > PR Reviewer) processing error logs in under 30 seconds end-to-end, eliminating manual debugging for 100% of tested workflow failures.

Enterprise Compliance Agent + LLMops Dashboard

Agentic Workflow | Python, FastAPI, Groq, GitHub Actions | Jan 2026 - Feb 2026

- Developed a stateful compliance agent with persistent memory and tool calling that processed 200+ document assessments with 97% domain-accuracy, blocking 100% of adversarial prompt injection attempts via pre-execution Input Guardrails.
- Built an automated LLM-as-a-Judge Evaluation Dashboard scoring agent responses across 5 compliance dimensions (accuracy, safety, relevance, completeness, tone) with 92% agreement with human evaluators.
- Implemented Dynamic Model Failover routing (70B > 8B) ensuring 99.9% uptime across 1,000+ API calls during rate-limit stress testing, with zero dropped requests.

PersonaDoc - Production RAG System

Retrieval-Augmented Generation | FastAPI, ChromaDB, Sentence-Transformers, LLaMA 3.3 | Nov 2025 - Dec 2025

- Architected a RAG pipeline that ingests enterprise PDFs (up to 500 pages) and returns cited, grounded answers in under 3 seconds, achieving 94% retrieval accuracy using semantic chunking and dense vector search with ChromaDB.
- Reduced LLM hallucination rate by 78% compared to raw prompting by implementing source-grounded generation with chunk-level citation tracking and confidence scoring across 3 REST endpoints (/upload, /search, /documents).

AI Code Reviewer + Meeting Intelligence System

LLM Microservices | Python, FastAPI, GitHub REST API, Groq | Sep 2025 - Oct 2025

- Built a code review microservice analyzing 100+ real GitHub PRs, detecting an average of 3.2 issues per review (bugs, security vulnerabilities, logic flaws) that standard static analysis tools missed, with strict JSON schema output.
- Developed a two-stage Meeting Intelligence pipeline processing 60-minute transcripts in under 8 seconds, extracting structured summaries, per-person action items with deadlines, and draft follow-up emails with 96% extraction

accuracy.

EDUCATION

Dr. Abdul Kalam Technical University

B. Tech in Electronics and Communication Engineering | Aug 2022 - May 2026 | CGPA: 7.9

CERTIFICATIONS

- Oracle GenAI Certified Professional (2025)
- Oracle Data Science Professional Certification (2025)
- IoT and Industrial Automation Certification (2025)